

Assignment #4: FP Iteration / Interpolation

Due: December 2, 2024 at 11:45 p.m.

This assignment is worth 10% of your final grade.

Warning: Your electronic submission on *MarkUs* affirms that this assignment is your own work and no one else's, and is in accordance with the University of Toronto Code of Behaviour on Academic Matters, the Code of Student Conduct, and the guidelines for avoiding plagiarism in CSCC37.

This assignment is due by 11:45 p.m. December 2. If you haven't finished by then, you may hand in your assignment late with a penalty as specified in the course information sheet.

- [15] 1. In A3 questions #4 and #5 you used the Fixed Point Theorem (FPT) to identify guaranteed intervals of convergence for the fixed point iteration $x_{k+1} = g(x_k)$. We introduced the FPT in lecture and outlined the proof, but we did not complete it. Now you will complete the proof.

- (a) State the FPT, carefully including the conditions and claim.
- (b) Repeat the first part of the proof, given in lecture, where we show that under the conditions of the theorem the iteration $x_{k+1} = g(x_k)$ converges to *some* point $\tilde{x}$ in the sink.
- (c) Prove that $\tilde{x}$ in (b) is not just *some* point in the sink, but is indeed a *fixed* point of the iteration.
- (d) Complete the proof by showing $\tilde{x}$ in (b) is *unique* in the sink.

(**Hint:** As with part (b), both parts (c) and (d) are best proven with the Mean Value Theorem (MVT); you just need to choose your variables carefully. Note that although there may be weaker arguments why (c) and (d) hold, for full marks you **must** use MVT when proving (c) and (d).)

- [15] 2. The equation $x^5 + 12x^3 - 130 = 0$ is known to have a real root near $x = 2$. Three iterative schemes are proposed to find this root:

- (i) $x_{k+1} = 130x_k^{-4} - 12x_k^{-1}$
- (ii) $x_{k+1} = \sqrt[3]{130/(x_k^2 + 12)}$
- (iii) $x_{k+1} = (4x_k^5 + 36x_k^3 + 130)/(5x_k^4 + 48x_k^2)$

- (a) Which of these schemes *could* be used to solve this equation, and why?
- (b) Which scheme *should* be used, and why?
- (c) Derive an even better scheme, and explain why it is better.

- [10] 3. Determine p , q and r so that the order of the fixed-point iteration

$$x_{k+1} = px_k + \frac{qa}{x_k^2} + \frac{ra^2}{x_k^5}$$

for computing $a^{1/3}$ (i.e., the cube root of $a \in \mathbb{R}$) becomes as high as possible. For this choice of p , q and r , indicate how the error in x_{k+1} depends on the error in x_k .

- [15] 4. Consider the data points $\{(-1, 4), (0, 6), (1, 12)\}$.

- Using the method of undetermined coefficients, derive the monomial basis form of the quadratic polynomial interpolating this data. You **do not** need to first compute the $PA = LU$ factorization of the Vandermonde matrix. Just solve the Vandermonde system for the coefficients of the monomial basis quadratic interpolant.
- Derive the Lagrange basis form of the quadratic interpolant. Show all of your work.
- Derive the Newton basis form of the quadratic interpolant by setting up and solving the lower-triangular system arising from the interpolation conditions. (Do **not** use divided-differences.)
- Verify, in two different ways, that the interpolants derived in (a), (b) and (c) are indeed the same polynomial. (**Hint:** One way is to convert the interpolants in (b) and (c) to monomial basis form and then compare to the interpolant in (a). The other, more elegant way, is to use an appropriate theorem discussed in lecture.)
- For a general set of $n + 1$ data points $\{(x_i, y_i), i = 0, 1, \dots, n\}$, what is the complexity of each approach in (a), (b), and (c)? Count flops (floating point operations). Note that you are counting flops incurred during the *construction* of each interpolant; you are not considering the cost of *evaluating* each interpolant.

- [5] 5. The monomial basis coefficients of

$$p(x) = \sum_{i=0}^n a_i x^i$$

and the Newton basis coefficients of

$$p(x) = \sum_{i=0}^n a_i \prod_{j=0}^{i-1} (x - x_j)$$

can each be computed by solving an $(n + 1) \times (n + 1)$ linear system. The monomial basis coefficients are computed by solving a dense Vandermonde system, as in #4(a), and the Newton basis coefficients *could* be computed by solving a lower-triangular system, as in #4(c), although in practice divided-differences are used.

Show that the coefficients of the Lagrange interpolant could also be computed by solving an $(n + 1) \times (n + 1)$ linear system. What is the floating point operation complexity of solving this system?

[total: 60 marks]